

interface to transmit monitored data.

The advantage is the easy monitoring of connected refrigeration units. In situations that need attention, the concerned person gets a notification instantly. Real-time data enables quick decisions, ultimately saving time and money, besides protecting consumers' health. Transparency for audits is also ensured as the entire operation of food

storage gets recorded.

Farmer's Fridge is another provider using IoT services from telecommunications company AT&T to deliver fresh food in vans to people on the go. Smart fridges are connected on a single scalable platform where the status is monitored to perform functions like automatic adjusting of temperature and generation of alerts

if maintenance is required. Apart from this, connected fridges are also capable of alerting drivers about their delivery route for the next day based on real-time data obtained from monitoring inventory and credit card transactions. Fridges get shut down automatically if there is a continuous power outage for two hours.

—Ayushee Sharma

4 MANY WAYS IoT-ENABLED SENSORS CAN HELP IN LOGISTICS

The introduction of the Internet of Things (IoT) in logistics industry can be beneficial not only for the companies but also for the consumers. Customers can track the shipment of their products and know the exact delivery date among other things

The Internet of Things (IoT) is playing a great role in the growth of several industries by providing real-time data and operational efficiency. Logistics is one of the primary sectors that is benefiting greatly from IoT.

IoT, in conjunction with emerging technologies such as Big Data, provides an efficient supply chain platform for the logistics industry through complete transparency and faster detection of issues. Inventory management and warehousing are the most important parts of a connected logistics system. IoT-enabled sensors allow companies to easily track inventory items, monitor their status, and create a smart warehouse system. By using this data, companies can even predict equipment failures and reduce the associated risks.

Further, IoT technology helps enhance the security of a warehouse. For example, CCTVs and alarm sensors use IoT to announce thefts and protect assets. Remote alarm monitoring can also help monitor video data at any given time across different locations.

While safety is important for warehouse management, most logistics companies find it challenging to ensure the safety of their employees. However, companies that are using such new technologies have managed to protect their staff by reducing acci-

dents at work to a great extent. With real-time data, companies can check the performance of their drivers and vehicles, which helps increase safety and reduce inventory damage.

As a tracking tool, IoT provides real-time tracking to enable customers to track their products. Companies like Amazon use radio frequency identification (RFID) tags that help customers track the shipment of their products. RFID tags have an integrated circuit that allows the customers to detect the location of their goods.

In addition, IoT-based sensing technology can maintain temperature stability and minimise the wastage of perishable products, thereby enhancing the end-to-end supply chain visibility. It can capture data related to environmental factors such as pressure, humidity, and light exposure.

US-based company Union Pacific uses IoT to monitor equipment and predict possible failures. The company uses sensors to monitor the operation of train wheels. As a result, Union Pacific has been able to reduce derailment risks, thereby curbing work-related accidents.

Many companies like those involved in shipping heavy equipment are responsible for emitting carbon footprints every year. IoT devices can allow these companies to analyse, monitor, and even manage

the carbon footprint at every level. Companies can identify the process causing higher carbon emission and take measures to make the process more environment friendly.

To achieve success and growth in logistics industry, using predictive analytics with IoT is a must. Such tools are used for laying out the shortest delivery routes, detecting early signs of problems in equipment, and reminding staff when it's time to replace equipment components. This results in increased productivity of warehouses, timely replacement of defective machinery parts, and prevention of accidents.

For instance, DB Schenker, a global logistics provider, has introduced the Decision Support Tool. This software is used by the company at different warehouse locations to simulate daily scheduling and processes to optimise operations. The company has also launched Industrial Data Space, a platform for secure data exchanges between companies who are using predictive analytics to enable predictive maintenance.

IoT is sure to transform the logistics industry in numerous ways. With the development of low-cost IoT networks and continually falling sensor prices in the coming years, large-scale IoT deployments in logistics will be possible. **EFY**

—Pallavi Singh